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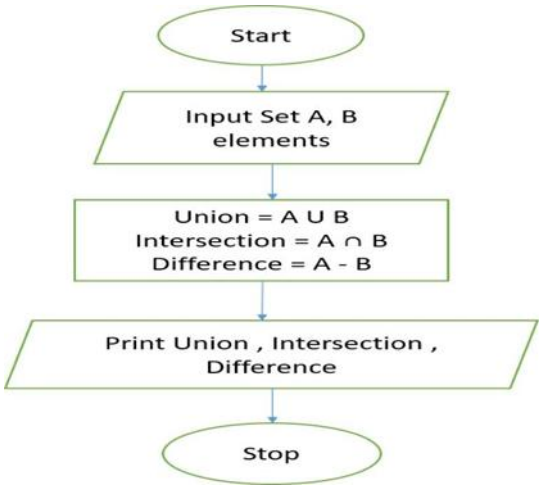
PPS 4.1.1

Algorithm: Set Operations

1. Start
2. Read space-separated integers for Set A.
3. Read space-separated integers for Set B.
4. Convert the inputs into two sets: Set A and Set B.
5. Find the Union , Intersection and Difference of Set A and Set B.
6. Display the Union, Intersection and Difference.
7. Stop

Flowchart:



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4.1.1. Set Operations

16:26

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer

setoperat...

Submit

Debugger

```python
input_a = input("Set A: ")
set_a = set(map(int, input_a.split()))
input_b = input("Set B: ")
set_b = set(map(int, input_b.split()))
union_result = set_a.union(set_b)
intersection_result = set_a.intersection(set_b)
difference_result = set_a.difference(set_b)
print(f"Union: {union_result}")
print(f"Intersection: {intersection_result}")
print(f"Difference: {difference_result}")
```

```
Set A: 1 2 3 4
Set B: 5 6 7 8
Union: {1, 2, 3, 4, 5, 6, 7, 8}
Intersection: set()
Difference: {1, 2, 3, 4}
=== YOUR PROGRAM HAS ENDED ===
```